

Homework: Linear Triangle Finite Element Method for an Unsteady Convection–Diffusion Equation

Numerical Methods for Partial Differential Equations

Problem Description

Consider the unsteady convection–diffusion equation on the unit square domain

$$\Omega = (0, 1) \times (0, 1), \quad 0 < t \leq T,$$

where $T = 1$. The governing equation is

$$\frac{\partial u}{\partial t} - \varepsilon \Delta u + \boldsymbol{\beta} \cdot \nabla u = f(x, y, t), \quad (x, y) \in \Omega, \quad 0 < t \leq T.$$

In this homework, use

$$\varepsilon = 0.01, \quad \boldsymbol{\beta} = (1, 1)^T.$$

Therefore, the PDE is explicitly given by

$$\boxed{\frac{\partial u}{\partial t} - 0.01 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = f(x, y, t)}$$

for

$$0 < x < 1, \quad 0 < y < 1, \quad 0 < t \leq 1.$$

The exact solution is prescribed as

$$u_{\text{exact}}(x, y, t) = e^{-t} \sin(\pi x) \sin(\pi y).$$

The initial condition is

$$u(x, y, 0) = u_{\text{exact}}(x, y, 0) = \sin(\pi x) \sin(\pi y), \quad (x, y) \in \Omega.$$

The boundary condition is

$$u(x, y, t) = u_{\text{exact}}(x, y, t), \quad (x, y) \in \partial\Omega, \quad 0 < t \leq 1.$$

Since

$$\sin(\pi x) \sin(\pi y) = 0 \quad \text{on } \partial\Omega,$$

the boundary condition is equivalently

$$u(x, y, t) = 0, \quad (x, y) \in \partial\Omega, \quad 0 < t \leq 1.$$

Error Computation

At the final time $T = 1$, compute the nodal error

$$e_i = U_i^{N_t} - u_{\text{exact}}(x_i, y_i, T).$$

The discrete L^2 -error is computed by

$$\|u_h - u\|_{L^2(\Omega)} \approx \sqrt{e^T M e}.$$

The H^1 -seminorm error is computed by

$$|u_h - u|_{H^1(\Omega)} \approx \sqrt{e^T K e}.$$

Spatial Convergence Test

Use the mesh sizes

$$N_x = N_y = 8, 16, 32, 64.$$

Let

$$h = \frac{1}{N_x}.$$

To observe spatial convergence, choose a sufficiently small time step, for example

$$\Delta t = h^2.$$

For linear triangular finite elements, the expected convergence rates are

$$\|u_h - u\|_{L^2(\Omega)} = O(h^2),$$

and

$$|u_h - u|_{H^1(\Omega)} = O(h).$$

Fill in the following table.

$N_x = N_y$	h	Δt	L^2 -error	L^2 -rate	H^1 -rate
8	1/8				
16	1/16				
32	1/32				
64	1/64				

The convergence rate is computed by

$$\text{rate} = \frac{\log(E_h/E_{h/2})}{\log 2}.$$

Temporal Convergence Test

To test temporal convergence, choose a sufficiently fine mesh, for example

$$N_x = N_y = 128.$$

Run the computation with

$$\Delta t = \frac{1}{10}, \frac{1}{20}, \frac{1}{40}, \frac{1}{80}.$$

Since the Crank–Nicolson method is second-order accurate in time, the expected temporal convergence rate is

$$\|u_h - u\|_{L^2(\Omega)} = O(\Delta t^2),$$

provided the spatial error is sufficiently small.

Fill in the following table.

Δt	L^2 -error	temporal rate
1/10		
1/20		
1/40		
1/80		

The temporal convergence rate is computed by

$$\text{rate} = \frac{\log(E_{\Delta t}/E_{\Delta t/2})}{\log 2}.$$

Report Requirements

Your report should include the following items.

- (1) Clearly derive the source term $f(x, y, t)$ by substituting the prescribed exact solution into the governing equation.
- (2) Derive and state the weak formulation of the problem.
- (3) Derive and present the Crank–Nicolson time-stepping formula for the semidiscrete finite element system.
- (4) Provide a spatial convergence study and summarize the results in a convergence table.
- (5) Provide a temporal convergence study and summarize the results in a convergence table.
- (6) Compare the observed convergence rates with the expected theoretical rates, and briefly discuss any discrepancies.
- (7) Submit your complete computer code. In addition to the computation and assembly of the finite element matrices and the implementation of the Crank–Nicolson time-stepping scheme, the code should also include the following components:

- (a) Generation of a structured triangular mesh on the unit square.
- (b) A plot showing the spatial convergence behavior.
- (c) A plot showing the temporal convergence behavior.
- (d) Plots of the numerical solution and the exact solution at the final time $T = 1$.